

Your Docker containers are running perfectly... but what happens when one crashes, traffic suddenly spikes, or you need hundreds of replicas?



Why Docker Alone Isn't Enough



- Docker runs containers.
- Kubernetes manages containers at scale.
- Solves automation, scaling, self-healing, and high availability.



What is Kubernetes (K8s)?



- An open-source container orchestration platform.
- Automates deployment, scaling, networking, and management of containers.



Why Kubernetes for Microservices?



- Automatic scaling.
- Self-healing.
- Rolling updates.
- Zero-downtime deployments.
- High availability.



Kubernetes Architecture



- Control Plane (Master Node)
- Worker Nodes
- Pods
- Services
- Deployments



What is a Pod, Deployment & ReplicaSets?



What is a Kubernetes Service?



Internal Service Discovery, Rolling Updates, Self-Healing & Auto Scaling



What is Helm?



Industry Best Practices

